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ARTIFICIAL INTELLIGENCE (SECJ3553)
SECTION 03

GROUP 2

PROJECT:
SENTIMENT-BASED CLASSROOM PULSE DASHBOARD

THEME:
SMART EDUCATION

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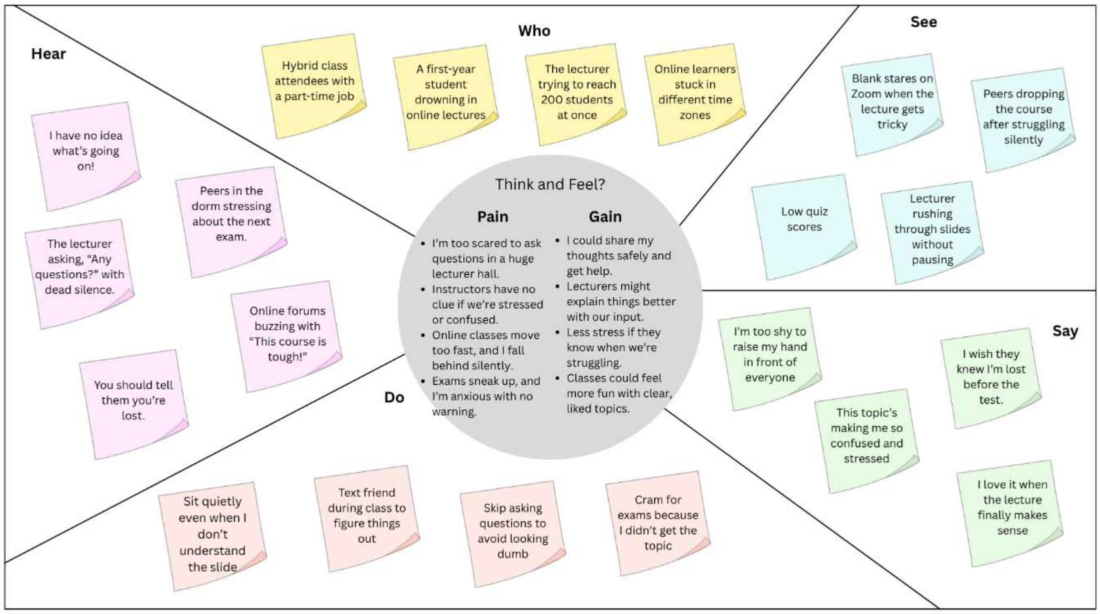
Table of Contents

Process of Emphasis in Design Thinking



We empathized with university students and lecturer by stating their classroom struggles in the sticky note sketch, inspired by comments from university Facebook Group and typical online learning challenges. The 'Pain and Gain' shows students' fear of speaking up and instructors' lack of insight, with gains like engagement. 'Who' lists first-years, online learners and lecturers, 'Hear' captures peer complaints, 'Do' notes silent confusion, 'See' spots rushed lecturers and 'Say' reflects stress like 'I'm lost!' This aims for a safe feedback tool to track sentiment fast.

Process of Define in Design Thinking

User	**Need**	**Insight**
University students including first-years, online learners, and hybrid students	Need 1: To figure out a safe and anonymous way to express confusion or ask for help during a large lecture.	First insight: Students are too scared and too shy to ask questions in front of everyone. This fear causes them to fall behind silently and sit quietly even when they do not understand the slide.
Lecturers (Managing large/online classes)	Need 1: To break the silence in class and get honest, real-time feedback on how well students understand the lesson. Need 2: Lecturer struggle with manual review of student surveys that are time-consuming and lack emotional depth. They want AI-based summaries to detect early signs of poor teaching effectiveness or student burnout.	First insight: End-of-semester surveys come too late to help. By the time lecturers read and analyse them, the students who were confused have already fallen behind. The feedback only shows what happened in the past, not something that can be used to fix problems right away. Second insight: A lecturer teaching 200 students cannot read everyone's comments while teaching. They need an AI assistant that quickly gathers and analyses anonymous feedback, then shows it on a simple visual dashboard. This helps the lecturer explain things better right away, instead of realizing the problem only after students have already fallen behind.

AI Solution

The proposed solution is a Sentiment-Based Classroom Pulse Dashboard, an AI-driven analytics platform designed to capture, analyse, and visualize student sentiment in real time. The system integrates with digital learning environments such as Moodle or Webex, automatically collecting textual data from student discussions, feedback forms, and chat interactions.

Using Natural Language Processing (NLP) and Sentiment Analysis, the AI model evaluates the emotional tone of student messages, categorising them into positive, neutral, or negative sentiments. Additionally, topic modelling techniques are used to identify the main themes associated with each sentiment cluster. This allows the system to pinpoint specific “emotional hotspots,” such as recurring frustration with certain topics such as assignment deadlines.

Goal of AI Solution

The main goal of the Sentiment-Based Classroom Pulse Dashboard is to enhance the quality of teaching and learning in Malaysian higher education by using AI to capture and understand student emotions in real time.

Through this system, the project aims to:

1. Support educators in identifying student frustration or disengagement early.
2. Improve student learning experiences by aligning teaching strategies with emotional and cognitive feedback.
3. Promote emotional well-being and inclusivity within the online platforms, especially for students who may be hesitant to express difficulties or opinions openly.
4. Empower universities with data-driven insights to improve course design and student support services.

Problem Statement

In Malaysian institutions, particularly in online or hybrid settings, instructors frequently find it difficult to gauge students' comprehension and emotions during lectures. Conventional feedback techniques, including end-of-semester questionnaires, do not adequately reflect the emotional context of the classroom and don't record student attitude in real time. Because of this, pupils who experience confusion, anxiety, or disengagement go overlooked until it is too late to take action.

Students, especially first-years, are reluctant to publicly express their perplexity because they are afraid of being judged or embarrassed. Unaddressed learning challenges, decreased engagement, and decreased overall teaching efficacy are the results of this communication gap.

An AI-driven system that automatically evaluates and visualizes classroom emotion is also required, enabling instructors to react quickly to indications of disengagement, bewilderment, or displeasure.